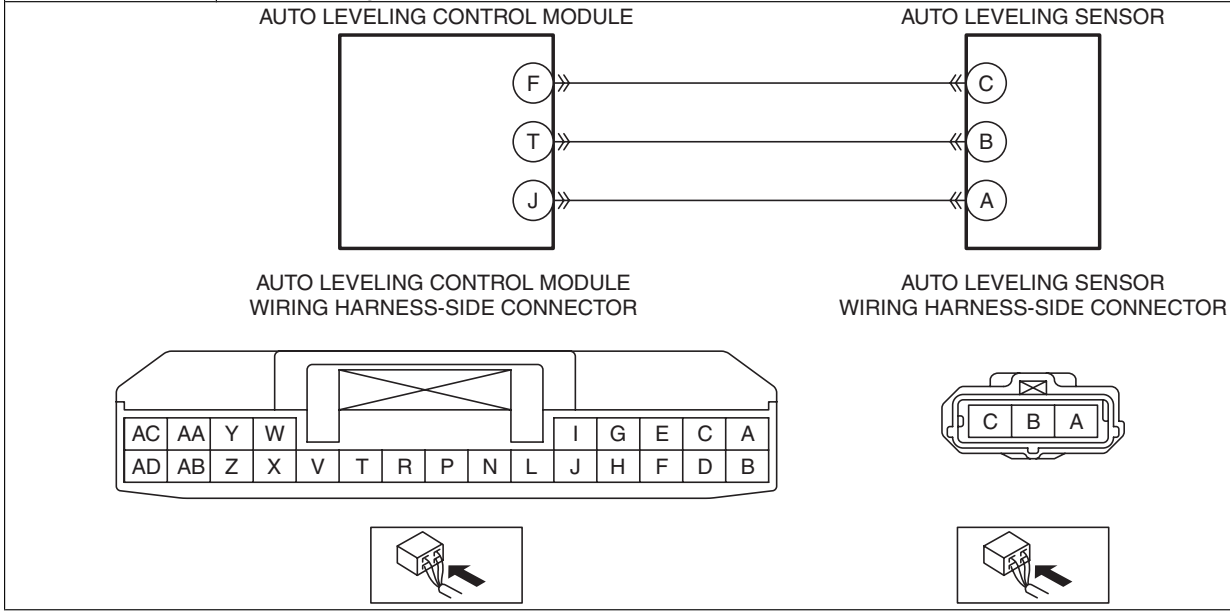
DTC B1044:01 [HEADLIGHT AUTO LEVELING SYSTEM]

id0902i7934200

Description	Auto leveling sensor circuit malfunction
Detection condition	• Auto leveling control module detected the auto leveling sensor circuit voltage of **0.25 V or less** or **4.75 V or more** for **10 s or more** with the ignition switched ON (engine off or on).
Fail-safe	• The headlight leveling actuator is stopped at the position when the malfunction is determined.
Possible cause	• Auto leveling sensor connector or terminal malfunction • Auto leveling control module connector or terminal malfunction • Short to ground in wiring harness between the following terminals: — Auto leveling control module terminal F—Auto leveling sensor terminal C — Auto leveling control module terminal T—Auto leveling sensor terminal B — Auto leveling control module terminal J—Auto leveling sensor terminal A • Short to power supply in wiring harness between the following terminals: — Auto leveling control module terminal F—Auto leveling sensor terminal C — Auto leveling control module terminal T—Auto leveling sensor terminal B — Auto leveling control module terminal J—Auto leveling sensor terminal A • Open circuit in wiring harness between the following terminals: — Auto leveling control module terminal F—Auto leveling sensor terminal C — Auto leveling control module terminal T—Auto leveling sensor terminal B — Auto leveling control module terminal J—Auto leveling sensor terminal A • Auto leveling sensor malfunction • Auto leveling control module malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT AUTO LEVELING SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the auto leveling sensor connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.

Step	Inspection		Action
2	**INSPECT AUTO LEVELING CONTROL MODULE CONNECTOR CONDITION** • Disconnect the auto leveling control module connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
3	**INSPECT AUTO LEVELING SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the auto leveling sensor and auto leveling control module connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Auto leveling sensor terminal C — Auto leveling sensor terminal B — Auto leveling sensor terminal A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Auto leveling control module terminal F—Auto leveling sensor terminal C • Auto leveling control module terminal T—Auto leveling sensor terminal B • Auto leveling control module terminal J—Auto leveling sensor terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
4	**INSPECT AUTO LEVELING SENSOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the auto leveling sensor and auto leveling control module connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side): — Auto leveling sensor terminal C — Auto leveling sensor terminal B — Auto leveling sensor terminal A • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Auto leveling control module terminal F—Auto leveling sensor terminal C • Auto leveling control module terminal T—Auto leveling sensor terminal B • Auto leveling control module terminal J—Auto leveling sensor terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 7.

Step	Inspection		Action
5	**INSPECT AUTO LEVELING SENSOR CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the auto leveling sensor and auto leveling control module connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Auto leveling control module terminal F—Auto leveling sensor terminal C — Auto leveling control module terminal T—Auto leveling sensor terminal B — Auto leveling control module terminal J—Auto leveling sensor terminal A • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Auto leveling control module terminal F—Auto leveling sensor terminal C • Auto leveling control module terminal T—Auto leveling sensor terminal B • Auto leveling control module terminal J—Auto leveling sensor terminal A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
6	**INSPECT AUTO LEVELING SENSOR** • Inspect the auto leveling sensor. (See AUTO LEVELING SENSOR INSPECTION.) • Is the auto leveling sensor normal?	Yes	Go to the next step.
		No	Replace the auto leveling sensor, then go to the next step. (See AUTO LEVELING SENSOR REMOVAL/INSTALLATION.)
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the auto leveling control module using the M-MDS. (See CLEARING DTC [HEADLIGHT AUTO LEVELING SYSTEM].) • Switch the ignition ON (engine off or on) and wait for **10 s or more**. • Retrieve the auto leveling control module DTCs. (See DTC INSPECTION [HEADLIGHT AUTO LEVELING SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [HEADLIGHT AUTO LEVELING SYSTEM].)
		No	DTC troubleshooting completed.